SACHIN J

EMBEDDED SYSTEMS | IoT | AI & AUTOMATION

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering student with a strong foundation in Embedded Systems, Internet of Things (IoT), and Artificial Intelligence. Proficient in C programming, ESP32, Arduino, sensor integration, and hardware-software interfacing. Experienced in developing innovative projects involving intelligent automation, IoT-based solutions, and AI-driven applications. A proactive and detail-oriented learner with strong problem-solving abilities, eager to apply technical knowledge and contribute to real-world engineering solutions.

TECHNICAL SKILLS

Programming	C, Embedded C, Python
Platforms	ESP32, Arduino; STM32 and Raspberry Pi (learning / project exposure)
Embedded & IoT	DC motor control, sensor interfacing, hardware-software integration, IoT concepts
Protocols & Tools	UART, I2C, SPI, MQTT (knowledge / exposure); Arduino IDE, Git, Keil, STM32CubeIDE
AI & Web	AI-assisted automation concepts, voice interaction, web application development, UI/UX

PROJECTS

Workspace AI Motor-Control Assistant | 2026 | ESP32, DC Motors, Motor Driver
- Tested forward and reverse motor rotation using an ESP32-based motor-control setup; voice-command integration for task-based movement is in development.

VECTOR – Personal AI Assistant | 2026 | Python, AI, Voice Interaction
- Developing a personal-assistant concept for conversational interaction and productivity support.

Car Accident Safety Monitor | 2026 | ESP32, Sensors, IoT
- Developing a sensor-based safety-monitoring concept for accident detection and emergency alerts.

FixNear – Service Marketplace Platform | 2025 | Web Development, UI/UX, Location-Based Services
- Designed a marketplace concept for nearby services and rentals, including category discovery and booking workflows.

AI Gesture and Voice-Controlled Drone | 2025 | Pixhawk, ESP32, GPS | Concept
- Designed a drone concept exploring voice/gesture interaction, GPS navigation, monitoring and safety alerts.

Autonomous Outdoor-Cleaning Robot | 2025 | Robotics, Motor Control, Sensors
- Developing a robotic-cleaning concept focused on automated movement, obstacle awareness and monitoring.

CERTIFICATIONS & TECHNICAL TRAINING

Robotics Internship — Yoi Robotics Laboratory LLP, Coimbatore (30 Sep – 16 Oct 2026)
Training areas: robot design and programming, circuit design, embedded systems, ROS 2, research papers and robot assembly.

Embedded Systems Design, Development & Testing — Krish Tec, Coimbatore (2 – 14 Jun 2025 | 45 hours)
Training areas: embedded hardware/software, microcontroller programming, IoT communication, PCB design, debugging and testing.

EDUCATION

B.E. Electronics and Communication Engineering — Kumaraguru College of Technology, Coimbatore (2022 – 2027)

Higher Secondary – Computer Science — Bharathi Vidya Bhavan Matriculation Higher Secondary School (2019 – 2022 | 83.1%)

LANGUAGES

Tamil (Native) | English (Fluent) | German (Basic)